HACKATHON 7 · EYES ON EARTH · SDG 15

HORNET CHECK

Asian hornet, or a harmless look-alike?

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Asian hornet in the Netherlands. Photo: Francina, iNaturalist (CC BY-NC)

THE PROBLEM

You see a big hornet in your garden. Report it, or leave it alone?

Invasive

The Asian hornet hunts honeybees, wild bees, butterflies and other insects.

Check first

Provinces ask you to confirm it is not a native European hornet. No photo, no report.

A wrong guess costs

In Limburg residents pay for nest removal themselves; Noord-Holland is considering it.

Sources: Provincie Drenthe (how to report the Asian hornet) · NH Nieuws, 18 Aug 2025

One invader, three look-alikes



Asian hornet

Dark body, one orange band, yellow leg tips

Report it



European hornet

Reddish-brown, mostly yellow, bigger

Native: leave it



Wasp

Small, bright black-and-yellow stripes

Common: leave it



Hornet hoverfly

A fly: two wings, cannot sting

Harmless

Photos via iNaturalist (CC BY-NC): Francina · Carel Jongkind · Rob Olivier · s_denheijer

OUR DATA

We scraped 700 Dutch photos from iNaturalist

600

to train in Roboflow, 150 per insect

100

kept apart, never seen by the model

385

different photographers

Our rules

- Observed in the Netherlands only
- Research grade: 2+ people agree on the species
- Open licence, with credits in attribution.csv
- One photo per observation
- Max 15 photos per photographer per insect

THE MODEL

From photos to a classifier in Roboflow

1

Scrape

Our Colab script downloads the photos

2

Upload

Folder names become the 4 labels

3

Prepare

224×224, flip and brightness ±15%

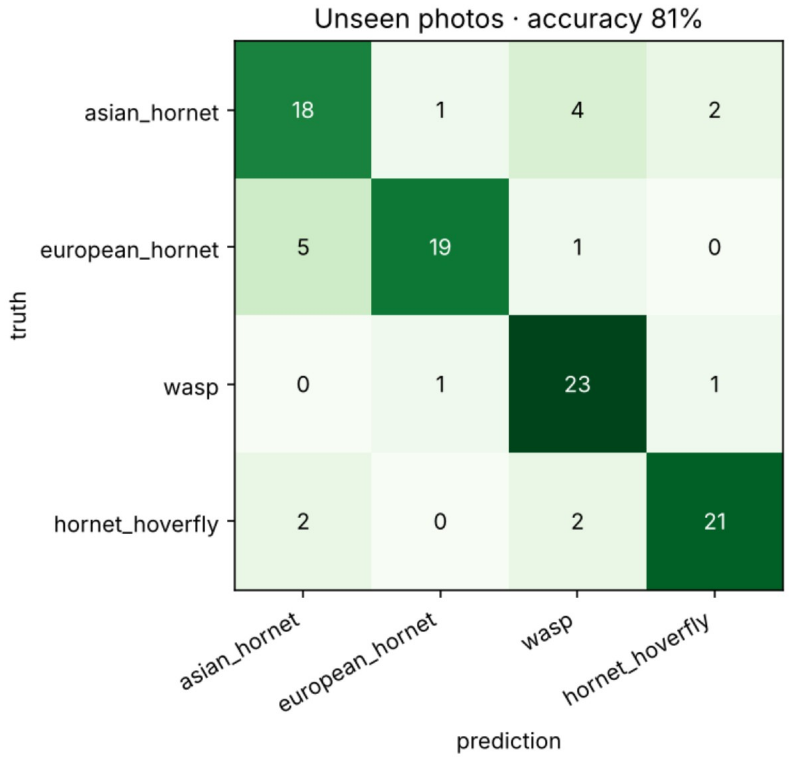
4

Train

DINOv3 (small) classification

Roboflow's own test split (60 photos): **precision 79.6% · recall 78.5% · F1 78.9%**

Training took about a minute; the screen estimated 3.5 hours.



OUR OWN TEST · 100 UNSEEN PHOTOS

81 / 100

correct overall

89%

correct when the app dares to give a verdict (81 photos)

18 / 25

Asian hornets recognised: the weakest class

We looked at every mistake



Several insects

Asian hornets plus a real wasp: “wasp”, 71%. One label cannot describe this photo.



Blur

A moving Asian hornet: “hoverfly”, 59%. Luckily the app says “not sure”.



Small and far away

A hornet as a dot on a wall: only 36% sure, so “not sure”.

Photos via iNaturalist: Herman Berteler (CC BY-NC) · groenteman (CC BY-NC) · vespawatch (CC0)

Designed around the worst mistake

A missed Asian hornet lets a nest grow unnoticed.
So the app:

> **Says “not sure” below 60%**
> instead of guessing

> **Warns at an Asian hornet score of 25%+**
> even when another insect wins

> **Lists features to check yourself**
> the human makes the final call

The price of the safety net

Warning at	Missed hornets caught	Extra warnings
10%	6 of 7	20
15%	4 of 7	13
20%	3 of 7	6
25% (ours)	2 of 7	3

We fixed 25% before testing. Picking the best limit on the same 100 photos would make our test look better than it is.

LIVE DEMO

LET'S TRY IT

Three photos the model has never seen, from our data/unseen folder.

1 · A clear Asian hornet

Red verdict and the advice to report it on waarneming.nl

2 · A hornet hoverfly

Harmless fly, nothing to report

3 · A hard Asian hornet

Model says "hoverfly" (45%), but the safety net warns: 37% Asian hornet

ETHICAL REFLECTION

A first check, not an official identification

What could go wrong

- A missed hornet: the nest grows unnoticed
- A false alarm: a native nest destroyed, money wasted
- Labels from volunteers, not experts
- Real phone photos are worse than our test photos
- It only knows four insects

What we did

- Never says “safe”, only “probably”
- “Not sure” and an Asian hornet warning
- Experts decide, via waarneming.nl
- Tells you to keep away from nests
- Credits and licences for all 700 photos

WHAT WE LEARNED

1. One score hides what matters: the most important insect was the weakest.

2. Looking at the wrong photos taught us more than the confusion matrix.

3. Decide which mistake is worse, and choose your limits before you test.

Next step: more photos of Asian hornets, and photos taken by ordinary phone users.